For Examiner's Use

1 A raindrop falls vertically from rest.

(a) Assume that air resistance is negligible.

On Fig. 1.1, sketch a graph to show the variation with time t of the velocity v of the raindrop for the first 1.0 s of the motion. [1]

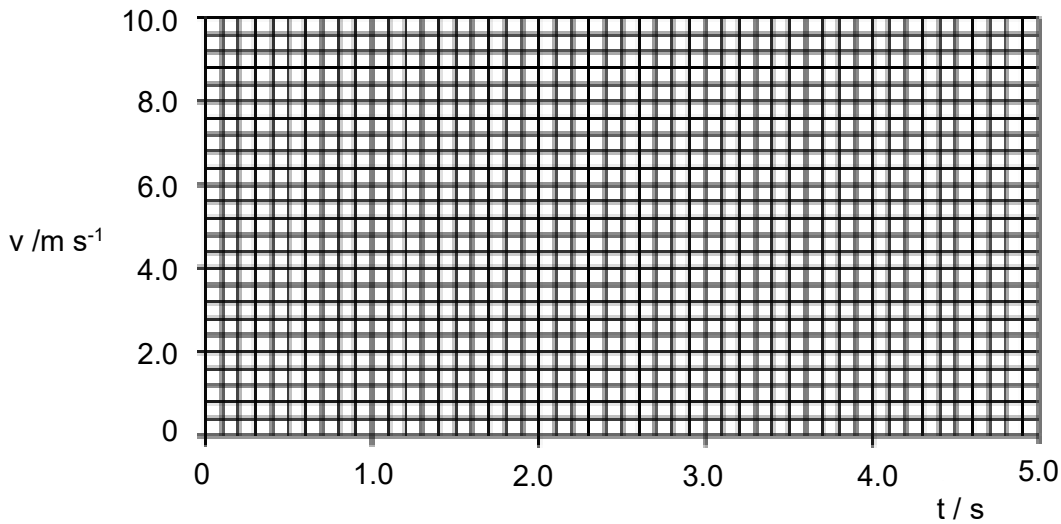

Fig. 1.1

(b) In practice, air resistance D on the raindrop is not negligible.

D is given by the expression

$$D = k\,v^2$$

where k is a constant and v is the speed.

(i) The raindrop has mass 1.38×10^{-5} kg and k is 2.76×10^{-6} N m^{-2} s^{2}.

Calculate the terminal velocity of the raindrop.

terminal velocity = …… m s^{-1} [2]

(ii) The raindrop reaches terminal velocity at t = 3.0 s.

On Fig. 1.1, sketch the variation with time t of velocity v for the raindrop. The sketch should include the first 5.0 seconds of the motion. [2]

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(c) A raindrop falls on a roof and rebounds off with a velocity of 5.5 m s^{-1} at an angle 60° with respect to the horizontal as shown in Fig. 1.2.

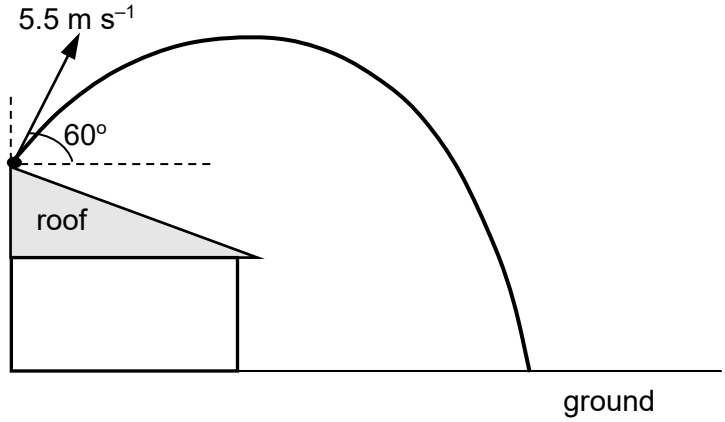

Fig. 1.2

Assume air resistance is negligible. The maximum horizontal distance travelled by the raindrop is 3.8 m.

(i) Calculate the time taken for the raindrop to hit the ground.

time = ……. s [1]

(ii) Determine the speed of the raindrop as it hits the ground.

speed = ……...m s^{-1} [2]

(iii) Discuss quantitatively whether the assumption that air resistance is negligible is justified by considering the vertical component of the initial velocity of the raindrop.

The raindrop has the same mass and dimension as in **(b)(i)**.

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…….…[2]